



# Where we are • what's new • today's plan

~25 min • recap • AI in the news • pre-read quiz • objectives & roadmap

## BY THE END OF THIS OPENING YOU WILL

Know where today sits in the course, what changed in the AI world this past week, and exactly what we will build — and break — in the next three hours.

## RECAP

# Where we were — Classes 1 & 2

*Two classes in one breath: the map of AI, and the system you build around a model.*

- 1 **AI is a taxonomy** — discriminative ML, generative, foundation models, agents, multimodal, physical AI
- 2 **The jagged frontier** — capability doesn't track human difficulty — you have to test it
- 3 **The AI stack** — compute → **embeddings** → **vector store** → model → serving → evals, layer by layer
- 4 **Build-vs-buy, per layer** — own-vs-rent judged on cost, latency, control, data gravity, regulation
- 5 **Cases — Moderna & DBS** — democratising AI across an org; a regulated bank's ADA and P-U-R-E governance
- 6 **You built** — a narrow model vs a foundation-model API; then **a RAG pipeline end to end**

**Throughline:** You learned the map, then built the system around the model.

## RECAP

## Where we were — Class 3

*You opened the model, learned to steer it, and proved the steering worked.*

- 1 **Inside the model** — tokens, attention, transformers, scaling laws — why capability emerges with scale
- 2 **Prompting craft** — zero/few-shot, chain-of-thought, self-consistency, structured JSON output
- 3 **Eval-driven iteration** — a 10-case harness, then climbing the prompt against it — measure, don't guess
- 4 **Live-coded build** — structured extraction; API and token-free local notebooks, run side by side
- 5 **Case — Med-PaLM 2 → AI Wars** — “it works — but can it pay?”, priced per token
- 6 **Guest lecture** — “Building your AI moats”: personalisation, safety/reliability, distribution, workflow

**Throughline:** You steered ONE call and proved it worked. Today you orchestrate a loop.

# AI in the news — agents in the wild

**THIS WEEK**

*Two developments from the last ten days — both about agents that act.*

**Agents ran an intrusion almost end to end.** Security firm Dream reported (12 Aug) a multi-agent system that took 2,500+ personnel records over four days — bypassing model guardrails by posing as an authorised penetration test.

**Why today:** these are Capsule 2's failure modes pointed the other way — and the clearest argument that permissions belong in code, not in a prompt.

**Agent-to-agent got a neutral home.** On 17 Aug, Google's Agent2Agent (A2A) moved under the Agentic AI Foundation at the Linux Foundation — alongside MCP, agentgateway and AGENTS.md, with 150+ partner organisations.

**Why today:** this is the plumbing under the word “tools” in Capsule 1 — and exactly the layer our guest speaker addresses.

Sources: [Dream Security](#) • [CyberScoop](#) • [CNN](#), 12 Aug 2026 • [Linux Foundation](#) • [Axios](#), 17 Aug 2026

**Takeaway** In one fortnight, agents were standardised and weaponised. Capability and control are arriving together.

# AI in the news — the economics

THIS WEEK

*The money story behind today's third capsule.*

| What changed                     | The number                                                                                                                                                       | Why it matters today                                        |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| The agent business got repriced  | Cognition — Devin's maker — in talks at ~US\$40bn, up from ~US\$25–26bn in May; the round is reported as tied to a US\$1bn annualised run-rate (US\$492m in May) | your Pre-read 5 company, two years after a 13.86% benchmark |
| Ambition far ahead of production | ~99% of firms say they plan agentic AI; only ~9–14% have it running in production                                                                                | the gap Capsule 3 explains — cost, reliability and trust    |

**The question this sets up:** everyone is buying agents; almost nobody is running them. What exactly is stopping the other 86%?

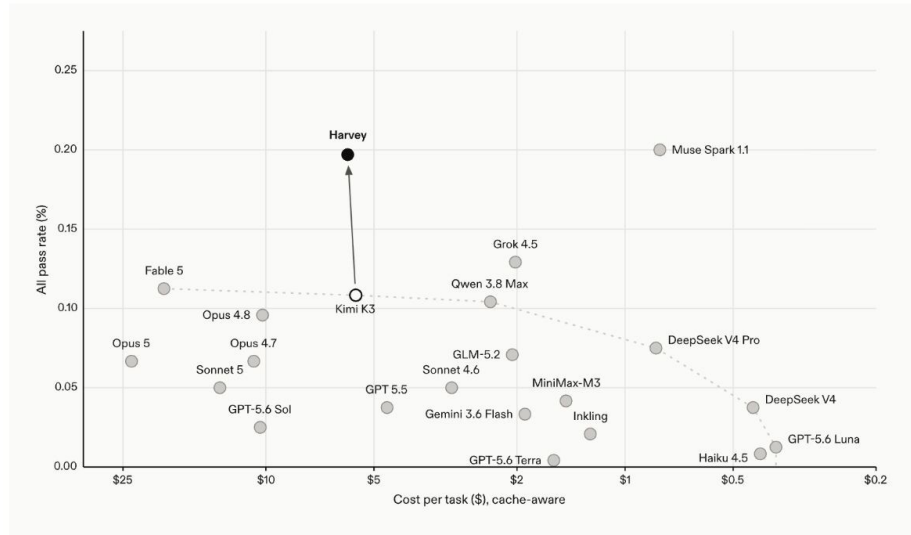
Sources: [TechCrunch \(Bloomberg\)](#), 12 Aug 2026 • [ANI adoption report](#), 19 Aug 2026

**Takeaway** Valuations price the promise; production rates price the reality. Today we work on the gap.

# AI in the news — a fine-tuned model, priced per task

THIS WEEK

Two ideas in one chart — and both of them land again in Capsule 3.



**Fine-tune for the task.** Harvey took a strong general model and tuned it on legal work: the all-pass rate roughly doubles — about 11% to 20% — at broadly the same cost per task. Specialisation, not scale.

**Price the TASK, not the token.** The horizontal axis is dollars per completed task, cache-aware — not price per million tokens. A cheaper token that needs more turns, retries and re-sent context can cost more per task.

~1,500 tasks • 24 practice areas • [Harvey Legal Agent Benchmark](#) (with Vals AI), Aug 2026

**Takeaway** Tune the model to the task — then price the task, not the token.

# What you read before today

*Six briefs — five foundations plus the case brief; Reading 3 travels with its Class 1 original. Each one seeds a capsule. Here's the through-line.*

- 1 **Anthropic — Building effective agents** — workflow or agent? the pattern ladder, and how to design tools → C1 • C2
- 2 **Yao — ReAct** — thought → action → observation: the loop you read in C1 and build in C2 → C1 • C2
- 3 **LeCun — objective-driven AI (revisited from Class 1)** — the dissent: no world model, no real planning → C1 • C3
- 4 **Hassabis — agents as a product** — an always-on agent, priced and permissioned → C1 • C3
- 5 **Cognition — Introducing Devin** — the claim, the independent field test, the correction → C2 • C3
- 6 **Case brief — Anthropic: From LLM to Agent?** — safety-first governance against the agent race → C3 (the case)

**Takeaway** Everything you read gets used today — the reading pays off in the room.

## Class 4 — objective & learning outcomes

*Grounded in the course outline, so you can see the line from today to assessment.*

### FROM THE OBTL+ OUTLINE • WEEK 4

Agents and tool use — agentic patterns (ReAct, planner-executor, multi-agent); tool design and integration; agent failure modes; agent economics; agent evals.

### THE COURSE'S FOUR INTENDED LEARNING OUTCOMES — AND WHERE TODAY MEETS THEM

|       | Upon successful completion of this course, you will be able to...                                                                                                                                             | Today                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| ILO 1 | Distinguish emerging AI technologies — discriminative ML, foundation models, retrieval-augmented, agentic, multimodal and physical AI — and select appropriate techniques for given business problem classes. | C1 — supporting         |
| ILO 2 | Build and validate an end-to-end AI solution to a defined problem using both low-code platforms and a code-based stack.                                                                                       | C2 — the build → A2     |
| ILO 3 | Evaluate design alternatives against business factors (cost, time-to-deploy, risk, ROI) and technical factors (accuracy, latency, scalability, complexity), and justify the chosen approach.                  | C3 — cost & reliability |
| ILO 4 | Communicate an AI solution — its problem framing, design choices, limitations, and risks — to a mixed technical and business audience.                                                                        | C3 — case discussion    |

**Takeaway** Today: build a loop, break it on purpose, price it — and learn when not to build one at all.

# Today's plan

*Where each piece happens — and roughly how long. One 15-min break, same for all three cohorts.*

|    | Block                                        | What happens                        | Min |
|----|----------------------------------------------|-------------------------------------|-----|
| C0 | Where we are • what's new • today's plan     | quiz • recap • AI news • objectives | ~25 |
| C1 | What is an agent?                            | lecture • poll • think-pair-share   | ~45 |
| C2 | Tools, failure modes & evals                 | lecture • quiz • live demo          | ~25 |
| —  | Break                                        |                                     | 15  |
| C2 | ...continued — build & break it              | code walkthrough • self-run         | ~30 |
| C3 | Agent economics + the Anthropic case         | case brief • discussion • share-out | ~45 |
| —  | Close — takeaways & what's coming in Class 5 |                                     | ~6  |

*~3:11 planned + buffer • A1 due Sun 30 Aug, 23:59. • Group A: guest lecture at 4:00 pm (~35 min) — timings compress.*

**Takeaway** Three capsules: what an agent is → build and harden it → what it costs.